Lesson 3: Linux File System Structure

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Project	My DevOps Journey 🚀

We'll learn how Linux stores and organizes files and folders. This is important because DevOps engineers regularly work with files, folders, configuration files and server directories.

1. First, understand the concept

In Windows, you have drives such as:

- C:\
- D:\
- E:\

Inside these drives, you create folders and files.

Linux organizes files differently. It uses one main starting directory called Root (`/`).

2. Important Linux directories

Directory	Simple meaning
`/`	Main starting directory
`/home`	Stores users' personal files
`/root`	Personal folder of the root user
`/etc`	Stores system settings

Directory	Simple meaning
`/var`	Stores changing data, such as logs
`/tmp`	Stores temporary files
`/bin`	Contains important commands
`/usr`	Contains many programs and utilities

Practical time!

First, check whether Ubuntu is installed.

1. Press the Windows key.
2. Search for `Ubuntu`.
3. If you see Ubuntu, open it.

If ubntu is not installed

Step 1: Open PowerShell as Administrator

1. Press the Windows key.
2. Search for PowerShell.
3. Right-click on Windows PowerShell.
4. Select Run as administrator.
5. Click Yes if Windows asks for permission.

Step 2: Check WSL

Copy this command:

```
wsl --status
```

Press Enter.

This command checks the status of WSL on your computer.

Step 3: Install Ubuntu

If WSL is not installed, execute:

```
wsl --install -d Ubuntu
```

Wait until the installation finishes. Windows might ask you to restart your laptop.

Step 4: Open Ubuntu

After installation:

1. Open the Start menu.
2. Search for Ubuntu.
3. Click Ubuntu.
4. Wait for the first-time setup.

It may ask you to create:

- UNIX username: Choose a simple username.
- Password: Create a password.

Important: When typing your Linux password, you won't see any characters on the screen. This is normal. Type the password and press Enter.

WSL stands for Windows Subsystem for Linux.

when Ubuntu has been installed successfully on your laptop.

Your output confirms:

- Virtual Machine Platform – Installed.
- Windows Subsystem for Linux (WSL) – Installed.
- Ubuntu – Installed.

There is just one step remaining.

Step 1: Restart your laptop

Your PowerShell says:

```
The requested operation is successful.
Changes will not be effective until the system is rebooted.
```

This means you must restart Windows before using Ubuntu.

Step 2: Open Ubuntu

After restarting:

1. Press the Windows key.
2. Search for Ubuntu.
3. Click to open it.
4. Wait for the first-time setup.

Step 3: Create your Linux account

Ubuntu will ask you to enter:

Username: Create a username (for example, `gaythri`).

Password: Create a password.

Important:

- Password characters won't appear while typing. This is normal.
- Remember your username and password.
- This is your Linux account, separate from your Windows account.

when Your Ubuntu installation is now successful!

You have completed your first DevOps practical setup.

tep 2: Our first Linux command

Once you see the prompt, type:

Press Enter.

What does `pwd` mean?

- `p` = Print
- `w` = Working
- `d` = Directory

It shows your current location in Linux.

if they ask allow Ubuntu to collect anonymous information to improve its features

then allow

Practical 1: Explore Linux directories

Now we'll perform our first three Linux commands.

Step 1: Check your current location

Type:

```
pwd
```

Press Enter.

What does it do?

It shows the folder where you are currently working.

Step 2: See the files and folders

Now type:

```
ls
```

This command displays the files and folders in your current location.

You might see no output. That's okay if your home directory is empty.

Step 3: Explore the main Linux directory

Now type:

```
ls /
```

This displays the main folders in Linux.

You may see:

```
bin  boot  dev  etc  home  lib  media
mnt  opt  proc  root  run  sbin  tmp
usr  var
```

These folders are part of the Linux file system.

Linux Practice – Day 1

Hands-on exercise

Execute these commands one by one:

```
pwd
ls
ls /
ls /home
```

Important: Execute each command separately and press Enter.

1. What does `/` mean?

`/` is called the Root Directory.

Think of it like the main folder of your computer. All other Linux folders and files are organised inside it.

Your output contains these folders:

Folder	Simple meaning
`/bin`	Important basic commands
`/boot`	Files needed to start Linux
`/dev`	Files connected to devices
`/etc`	System settings and configuration
`/home`	Normal users' personal folders
`/root`	Home folder of the root user (administrator)
`/tmp`	Temporary files
`/usr`	Many programs and system tools

Folder	Simple meaning
`/var`	Logs and files that keep changing
`/mnt`	Places where other drives or filesystems can be accessed
`/opt`	Additional software
`/proc`	Information about running processes and the system
`/sys`	Information about hardware and the kernel

Important: `/` and `/root` are different.

- `/` = Main directory of Linux.
- `/root` = Personal folder of the administrator account.

2. Let's do the next practical

Now let's check your personal folder.

Run this command:

```
ls /home
```

This command lists the folders inside `/home`.

Then run:

```
ls /home/Gaythri26
```

This shows the files and folders inside your Ubuntu user's home directory.

Let's create your first Linux folder!

Now we'll do a small practical task.

Step 1: Create a folder named `DevOps`.

```
mkdir DevOps
```

`mkdir` means Make Directory. It creates a new folder.

tep 2: Check whether the folder was created.

```
ls
```

Expected output:

```
DevOps
```

Next topic: `cd` – Change Directory

`cd` is used to move from one folder to another.

For example, you are currently here:

```
/home/Gaythri26
```

Now let's enter your `DevOps` folder.

Practical 1: Run:

```
cd DevOps
```

Practical 2: Check your current location:

```
pwd
```

Expected output:

```
/home/Gaythri26/DevOps
```

Remember these three commands:

Command	Meaning
`mkdir`	Create a folder
`ls`	List files and folders
`cd`	Move between folders

Next practical: Move one step back

Linux has a command called `cd ..`.

- `cd` = Change Directory
- `..` = Previous (parent) directory

Run:

```
cd ..
```

Then run:

```
pwd
```

Expected output:

```
/home/Gaythri26
```

Remember:

- `pwd` → Shows your current location.
- `cd ..` → Moves one folder back.
- `cd DevOps` → Enters the DevOps folder.

Next practical: Learn `cd ~`

There is another useful command:

```
cd ~
```

It takes you directly to your home directory, no matter where you are.

Let's test it.

Step 1: Enter your DevOps folder:

```
cd DevOps
```

Step 2: Return to your home directory:

```
cd ~
```

Step 3: Check your location:

```
pwd
```

Expected output:

```
/home/Gaythri26
```

Let's check what you did:

Command	What happened
`pwd`	Displayed your current location
`cd ~`	Moved to your home directory
`cd DevOps`	Entered the DevOps folder
`cd ~`	Returned to your home directory

Quick revision before the next topic

Try answering these three questions:

1. Which command displays your current directory?

ls

pwd

mkdir

2. Which command creates a new folder?

cd

ls

mkdir

3. What does cd .. do?

Creates a folder

Moves to the parent folder

Deletes a folder

Today we'll learn 5 important Linux commands through practical exercises inside your `DevOps` folder.

These commands are very important for DevOps because we regularly create, read, copy, move and delete files.

Linux practical: File management commands

First, understand what each command does:

Command	Meaning	Use
`touch`	Create a file	Create an empty file
`cat`	Display file content	Read a file
`cp`	Copy	Copy a file
`mv`	Move	Move or rename a file
`rm`	Remove	Delete a file

We'll practise one command at a time.

Practical 1: `touch` – Create a file

What is `touch`?

The `touch` command creates a new empty file.

Imagine you want to create a file named `notes.txt` inside your DevOps folder.

Step 1. First, enter your DevOps folder:

```
cd ~/DevOps
```

Step 2. Create a file:

```
touch notes.txt
```

Step 3. Check whether the file was created:

```
ls
```

Expected output:

```
notes.txt
```

ou have created your first file: `notes.txt` inside the `DevOps` folder.

Let's understand your commands:

```
cd ~/DevOps
```

Moved you inside the DevOps folder.

```
touch notes.txt
```

Created an empty file named `notes.txt`.

```
ls
```

Displayed the file:

```
notes.txt
```

Practical 1 completed: touch

Practical 2: `cat` – Read file content

`cat` stands for Concatenate. For now, we'll use it to display the contents of a file.

Your `notes.txt` file is empty. So first, let's add some text using another command.

Step 1: Write something into your file

Run:

```
echo "I am learning Linux for DevOps." > notes.txt
```

Simple meaning:

- `echo` displays text.
- `>` sends that text into a file.
- `notes.txt` is the file where the text will be saved.

Step 2: Read the file

Now run:

```
cat notes.txt
```

Expected output:

```
I am learning Linux for DevOps.
```

Important: `>` replaces the existing file content. We'll learn how to add more lines without replacing the old content later.

1. `echo` command

```
echo "I am learning Linux for devops" > notes.txt
```

- `echo` → Displays text.
- `>` → Sends the text into a file (overwrites existing content).
- `notes.txt` → Destination file.

2. `cat` command

```
cat notes.txt
```

Displays the content of the file.

Next: `cp` – Copy a file

Imagine you have `notes.txt` and want to create another file with the same content.

We use `cp` (Copy).

Step 1: Run:

```
cp notes.txt backup.txt
```

This creates a new file named `backup.txt` with the same content as `notes.txt`.

Step 2: Check both files:

```
ls
```

Expected output:

```
backup.txt  notes.txt
```

Step 3: Read the copied file:

```
cat backup.txt
```

Expected output:

```
I am learning Linux for devops
```

Remember:

```
cp source destination
```

- Source = Original file.
- Destination = New copied file.

Let's verify your copied file

Run these two commands:

Step 1:

```
ls
```

Expected output:

```
backup.txt  notes.txt
```

Step 2:

```
cat backup.txt
```

Expected output:

```
I am learning Linux for devops
```

Practical 4: `mv` – Move and Rename files

`mv` means Move.

It has two important uses:

1. Move a file from one folder to another.
2. Rename a file.

Let's first practise renaming.

Step 1: Rename backup.txt

Run:

```
mv backup.txt old_notes.txt
```

Step 2: Check the files

Run:

```
ls
```

Expected output:

```
notes.txt  old_notes.txt
```

Step 3: Read the renamed file

Run:

```
cat old_notes.txt
```

Expected output:

```
I am learning Linux for devops
```

Important difference:

Command	Result
`cp notes.txt backup.txt`	Creates a copy; original remains
`mv backup.txt old_notes.txt`	Renames the file; old filename disappears

Try these three commands and share your output. Then we'll practise moving files between folders.

Next: Practise moving a file into another folder

Now let's understand the second use of `mv`.

Step 1. Create a folder named `Backup`:

```
mkdir Backup
```

Step 2. Move `old_notes.txt` into that folder:

```
mv old_notes.txt Backup/
```

Here:

- `old_notes.txt` = File to move.
- `Backup/` = Destination folder.

Step 3. Check the current folder:

```
ls
```

Expected output:

```
Backup  notes.txt
```

Step 4. Check inside Backup:

```
ls Backup
```

Expected output:

```
old_notes.txt
```

One final command: `rm`

`rm` means Remove. It deletes files.

Since you want to avoid learning too much in one day, we'll do just one small practical.

You currently have two files inside `backup`.

Run:

```
rm backup/old_notes.txt
```

Then check:

```
ls backup
```

Expected output:

```
notes.txt
```

Important: `rm` deletes files. Be careful because deleted files may not be easy to recover. For now, delete only the practice file specified above.